Subject Description Form

Subject Code	ITC6860
Subject Title	Technology II : Methods in Garment Pattern Engineering
Credit Value	3
Level	6
Pre-requisite / Co-requisite / Exclusion	Nil
Objectives	This course provides students with solid background in the theoretical and practical application of garment pattern engineering methods in solving manufacturing, sizing and management problem in the fashion industry. There are two sections: general topics and special topics. For the general topics, the main objectives are: (1) to train up the students to understand theory and techniques in garment pattern engineering and the related concepts, such as body scanning and body sizing; (2) to apply the garment pattern engineering techniques to improve the manufacturing; (3) to apply the concept of body scanning in creation of new products and services; (4) to apply the body sizing techniques to improve the customer-specific fitting and over all fitting of a target consumer market; (5) to evaluate the techniques and commercial systems. Student also needs to select a special problem, to study the topic in depth, to select the right method and tools to solve the problem. Students are required to attend seminars, read assigned readings, complete a project, and an examination.
Intended Learning Outcomes	Students who have successfully completed the subject should be able to: a. Understand the theory of the garment pattern engineering methods; b. Derive the mathematical model for the problem; c. Select the right computational methods to solve a problem; d. Apply theory and methods in solving a problem; e. Evaluate the performance and limitations of the garment pattern engineering methods.

Subject Synopsis/ Indicative Syllabus	This subject aims to provide students with theoretical and professional knowledge in advanced garment engineering methods in solving industrial problems. The syllabus is divided into FOUR main parts: Part 1: Introduction to mathematical modeling - Introduction to the translation of the daily problem into a mathematical format; - Understanding the importance of the definition of terms and concepts; - Expressing the mathematical equation of the problem in a proper format. Part 2: Introduction to garment pattern engineering methods - Different methods of body scanning and automatic body measurement. - Different methods of garment pattern unfolding. - Different methods of mark lay-planning. - Different methods of virtual garment sewing. - Different methods of virtual garment visualization. - Different methods of garment sizing generation. - Artificial intelligent methods. Part 3: Analysis of computational methods - Introduction to the performance evaluation, such as speed, scalability, accuracy, etc.; - Introduction to the theoretical limitations of the computational methods; - Comparison of different computational methods. Part 4: Selection of software tools - Introduction to the selection of software for the computation methods; - Performance benchmarking
Teaching/Learning Methodology	Students need to select a particular class of garment pattern engineering problem, such as pattern unfolding or soft computing (artificial intelligence), and pursue an independent study of the subject. This subject comprises lectures and seminars. Theories and concepts of different basic computational methods are conveyed by lectures. Seminars are arranged to deepen students' knowledge on the advanced garment pattern engineering methods and theoretical consideration when applying the methods in real world problems. Group/individual projects and assignments are employed as the assessment tools for students' strategic thinking, team working and presentation skills.

Assessment Methods in Alignment with Intended Learning Outcomes	(see table below)

Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed				
		a	b	c	d	e
1. Presentation	25%	✓	✓			
2. Project	20%		✓	✓	✓	✓
3. Examination	55%	✓	✓	✓	✓	✓
Total	100%					

Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes:

- Group/individual projects and assignments are employed as the assessment tools for students' strategic thinking, team working and presentation skills.

- Examination is the integration of their understanding, ability to formulate and handle the problem.

Student Study Effort Expected		
	Class contact:	
	▪ Lecture	10 Hrs.
	▪ Student presentation	7 Hrs.
	▪ Seminars	9 Hrs.
	Other student study effort:	
	▪ Reading	69 Hrs.
	▪ Project	25 Hrs.
	Total student study effort	120 Hrs.

Reading List and References	**<u>Indicative Reading List and References (e-book available in PolyU library)</u>:** 1. Soft computing in textile engineering [electronic resource]. Majumdar. Woodhead Publishing, 2011. 2. Kansei engineering and soft computing [electronic resource]: theory and practice / Ying Dai, Basabi Chakraborty, and Minghui Shi, editors. Hershey, Pa.: IGI Global, 2011. 3. Sizing in Clothing [electronic resource]: Woodhead Publishing Online. Woodhead Publishing in association with The Textile Institute; CRC Press, 2007. 4. Curves and surfaces for computer-aided geometric design : a practical guide, Farin, Gerald E, Academic Press , c1997 , 4th (Circulation Coll T385 .F37 1997) 5. Computer Modelling for Garment Pattern Design, Ng, R. PhD Thesis, Hong Kong Polytechnic University, 1998. (LG51 .H577P ITC 1998 Ng) **<u>Articles</u>**: 1. Yu Chen, Xianyi Zeng, Michel Happiette, Pascal Bruniaux, Royer Ng, Winnie Yu,"A new method of ease allowance generation for personalization of garment design", International Journal of Clothing Science and Technology, 2008, Vol. 20, No. 3, page 161-173. 2. Ng, R., Ashdown, S.P, & Chan, A., (2007) 'Intelligent size table generation.' Proceedings of the Asian Textile Conference (ATC), 9th Asian Textile Conference, Taiwan 3. NG, Roger, YU, Winnie, NG, S.P., "Intelligent Surface Model of Anisotropic Woven Tensile Property with ANN", 4th Autex Conference, Roubaix, 22-24th June 2004, ISBN 2-9522440-0-6, article ID P-SIM6. 4. Roger NG, Carine WONG, "Designing Garment Pattern With Artificial Neural Network", CESA 2003, July 7-9 2003, Lille, France, article ID: S4-I-01-0365. [Note: additional reading varies according to the special topics to be selected by the students] **<u>Journals</u>**: Clothing And Textiles Research Journal International Journal Of Clothing Science And Technology Journal Of Textile Institute Research Journal Of Textile And Apparel Sen'i Gakkaishi Taiwan Textile Research Journal Textile Research Journals